# SALVADOR CATSIS

**Computational Scientist · Mathematical Modelling & Simulation**

London, United Kingdom
https://linkedin.com/in/linkedin.com/in/salvador-catsis-b91956161
Contact details on request

## PROFILE

Computational scientist with seven years' experience building and analysing mathematical models of real systems, from NHS treatment pathways to physical and biological processes. First author on three peer-reviewed papers, co-author on three others, and co-inventor on two US patents.
I enjoy developing mathematical models of real-world systems and then building software to simulate them. I'm particularly interested in how far a model's predictions hold once the real system stops behaving ideally. Looking for part-time or contract work in modelling, simulation and data analysis.

## EXPERIENCE

### PhD Researcher

**University of Bristol**

Feb 2022 – Present — Bristol, UK

- Research on the sensitivity of nonsmooth dynamical systems to real-world imperfections — whether an idealised model stays reliable when a real system cannot switch instantaneously.
- Analysed invariant sets of one-dimensional systems with a single switch, using exact rational and interval arithmetic for rigorous bounds, alongside mathematical optimisation (JuMP/HiGHS) and symbolic computation.
- Built and maintained a GPU-accelerated simulation and analysis library (∼11,000 lines, Julia/CUDA) for studying the discretisation of two-dimensional nonsmooth systems with two switches.
- Two papers in *Physica D: Nonlinear Phenomena*, with a third in preparation.

### Research Associate

**University of Bristol**

Aug 2021 – Jan 2022 — Bristol, UK

- Modelled NHS cardiovascular referral, diagnostic and treatment pathways in England to help commissioners address COVID-19 waiting-list backlogs.
- Automated acquisition of routinely collected data from NHS Digital, NHS England, ONS and StatsWales (Python, Selenium), then implemented a system dynamics model (PySD, ∼6,900 lines) and calibrated it against that data using constrained optimisation (SciPy), fitting on a pre-pandemic window and projecting forward to test predictive performance.
- Identified restricted primary-care referrals as the dominant bottleneck behind a 40%+ rise in waiting lists; presented findings to NHS commissioners. First author of the resulting *BMJ Open* paper, coordinating ten co-authors across cardiology, surgery, operational research, mathematics and computing.
- Named contributor to a discrete-event simulation (SimPy) of the chronic heart failure diagnosis pathway, restructuring an inherited prototype into a documented six-module package.

### Doctoral Researcher (Causal Inference)

**Universitat de València**

Nov 2020 – Mar 2021 — Valencia, Spain

- Machine learning research at the Image Processing Laboratory under Prof. Gustau Camps-Valls, developing and benchmarking methods for discovering causal direction from observational data.

## LANGUAGES

Python · Julia · MATLAB · OpenCL · CUDA · Bash · LaTeX

## PYTHON

pandas · NumPy · SciPy · scikit-learn · TensorFlow · JAX · OpenCV · networkx · SimPy · PySD · Matplotlib · Selenium · Sphinx · Jupyter

## SIMULATION

System dynamics · Discrete-event simulation · Agent-based modelling · Stochastic simulation · Markov chains · Epidemic models · Network science · Model calibration · Forecasting & scenarios · Sensitivity analysis

## ANALYSIS

Classification · Ensemble methods · Kernel methods · Neural networks · Causal inference · PCA · t-SNE · Cross-validation · Hyperparameter tuning · Optimisation · Information theory

## COMPUTING

GPU computing · HPC clusters · Git · Linux · Package design · Technical documentation · AI-assisted development

## PUBLICATIONS

- Built three approaches in Python: a kernel-based ensemble using support measure machines; an information-theoretic method based on minimum description length; and four variants of the NOTEARS algorithm for causal graph structure learning. Benchmarked against established baselines on the Tübingen cause-effect pairs database and on simulated data with known ground truth (scikit-learn, pandas, NumPy, SciPy, JAX).
- Contributed conceptualisation, methodology, software and the original draft of the resulting paper in *Applied Soft Computing*.

### Research Engineer

**Ultraleap**

Oct 2018 – Sep 2019 Bristol, UK

- Developed algorithms for controlling phased-array ultrasound to render tactile sensations in mid air, formulated as an inverse problem for the transducer activations producing a desired acoustic field.
- **First-named inventor** on the resulting patent, contributing two solution methods: singular value decomposition with Tikhonov regularisation, and mini-batch stochastic gradient descent with momentum.
- Named author of `handybeam`, Ultraleap's open-source GPU-accelerated acoustic simulation package (Python, OpenCL), as primary contributor of the coordinate translator and two samplers.
- Reported progress quarterly to an Innovate UK funding officer in written reports and presentations.

### Undergraduate Researcher

**Bristol Robotics Laboratory**

Jun 2017 – Aug 2017 Bristol, UK

- Wrote the optical tracking software (Python) and the Arduino firmware controlling the experimental rig for artificial cephalopod skin, within a larger codebase led by the project's lead researcher. Co-author on two conference papers.

### LMS Undergraduate Research Student

**University of Bristol**

Jun 2016 – Aug 2016 Bristol, UK

- Modelled SIS epidemics on spatial random geometric graphs to predict which individuals are most likely to be infected. Competitively awarded bursary from the London Mathematical Society.
- Used principal component analysis on simulation output to establish how many network properties predict a node's steady-state infection probability, then developed and compared three analytical models, identifying where each broke down.

## EDUCATION

### PhD in Engineering Mathematics

**University of Bristol**

2022 – Present

Sensitivity to perturbations in non-smooth dynamical systems, combining analytical and numerical methods.

### MEng Engineering Mathematics (First Class, 81%)

**University of Bristol**

2014 – 2018

**Process modelling of NHS cardiovascular waiting lists in response to the COVID-19 pandemic.** *BMJ Open* 13(7), 2023. *First author.*

**Perturbation and determinacy of nonsmooth systems.** *Physica D* 477, 2025. *First author.*

**The hidden sensitivity of non-smooth dynamics.** *Physica D* 463, 2024. *First author.*

**Pairwise causal discovery with support measure machines.** *Applied Soft Computing* 150, 2024.

Two conference papers on artificial cephalopod skin: *SPIE EAPAD XX* and *IEEE RoboSoft*, 2018.

## PATENTS

**Ultrasound acoustic field manipulation techniques.** US 2022/0252550 A1. *First-named inventor.*

**Features of airborne ultrasonic fields.** US 11,816,267 B2, 2023.

## DOMAINS

- Healthcare systems
- Climate & energy
- Haptics & acoustics
- Robotics
- Applied dynamics

## AVAILABILITY

2–3 days per week.
Permanent, part-time or contract.
London or remote.